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Cell-o1: training LLMs to solve single-cell reasoning puzzles with reinforcement learning

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Abstract

Motivation: **Large language models (LLMs) have demonstrated strong general reasoning abilities, but applying them to domain-specific tasks such as analysing single-cell RNA sequencing data remains a challenge. A central task in this domain is cell type annotation, which is critical for understanding cellular heterogeneity. Although recent foundation models attempt to automate this process, they typically annotate cells independently, without considering batch-level context or providing explanatory reasoning. To address this limitation, we introduce the `CellPuzzles` benchmark, which reformulates cell type annotation as a batch-level reasoning task. `CellPuzzles` spans diverse tissues, diseases, and donor conditions, and requires reasoning across the batch-level cellular context to ensure label uniqueness.**

Results: **We find that off-the-shelf LLMs struggle on this task, with the best baseline (OpenAI `o1`) achieving only 19.0% batch-level accuracy. To fill this gap, we propose `Cell-o1`, a 7B LLM trained via supervised fine-tuning on distilled reasoning traces, followed by reinforcement learning with batch-level rewards. `Cell-o1` achieves state-of-the-art performance, outperforming OpenAI `o1` by over 73% and generalizing well across contexts. Further analysis of training dynamics and reasoning behaviors provides insights into batch-level annotation performance and emergent expert-like reasoning.**

Availability and Implementation: **Code and data are available at https://github.com/ncbi-nlp/cell-o1.**

Introduction

Assigning accurate cell types to single-cell RNA sequencing (scRNA-seq) profiles is fundamental for understanding biological heterogeneity across tissues, diseases, and individuals (Kiselev et al. 2019, Luecken and Theis 2019). Traditional annotation workflows rely heavily on expert knowledge, typically involving clustering to group similar cells (Butler et al. 2018, Traag et al. 2019), followed by manual inspection of marker gene expression to assign cell type labels based on biological domain knowledge (Aran et al. 2019, Stuart et al. 2019). While accurate, this process is time-consuming and labor-intensive, with limited scalability across large or new datasets (Abdelaal et al. 2019, Kiselev et al. 2019, Luecken and Theis 2019, Stuart et al. 2019, Lähnemann et al. 2020).

Recent advances in deep learning have significantly improved automated cell type annotation through the development of single-cell foundation models (Yang et al. 2022, Theodoris et al. 2023, Hao et al. 2024). These models leverage large-scale unsupervised pre-training to capture complex gene-expression patterns, enabling better representation learning and enhanced performance across various downstream tasks. In parallel, large language models (LLMs) have also been adapted for single-cell applications, either by translating gene-expression profiles into textual representations (Liu et al. 2023, Hou and Ji 2024, Levine et al. 2024, Chen and Zou 2025) or by integrating multimodal gene embeddings (Schaefer et al. 2024, Zhao et al. 2024, Fang et al. 2025, Shi et al. 2025). However, both foundation models and LLMs typically annotate each cell independently without considering the shared biological context or batch-level gene-expression information, which fundamentally differs from expert annotation practices (Pasquini et al. 2021). Furthermore, most automated methods directly predict cell types without articulating the underlying reasoning process, making their decisions difficult to interpret and validate (Azodi et al. 2020).

To bridge this gap, we first introduce `CellPuzzles`, a novel benchmark that formulates cell type annotation as a batch-level reasoning task, closely mimicking expert annotation workflows.

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As illustrated in Fig. 1, unlike traditional methods that label each cell independently, `CellPuzzles` requires jointly assigning unique labels to all cells in a batch, using both their gene-expression profiles and shared contextual metadata. Extensive evaluations show that state-of-the-art LLMs struggle with this formulation, with the best-performing model [OpenAI `o1` (Jaech et al. 2024)] achieving only 19.0% batch-level accuracy, reflecting the complexity of this task.

To address this challenge, we propose `Cell-o1`, a reasoning-enhanced LLM trained in two phases as shown in Fig. 1: supervised fine-tuning (SFT) on expert-like reasoning traces distilled from a frontier LLM to guide structured and interpretable decision-making; followed by reinforcement learning (RL) with batch-level rewards to encourage consistent, context-aware label assignments. On `CellPuzzles`, `Cell-o1` outperforms all baseline models in both cell-level and batch-level accuracy. Further analyses of training dynamics, reasoning behavior, and prediction errors offer insights into the model's generalization, interpretability, and reasoning ability. Notably, `Cell-o1` exhibits emergent behaviors such as self-reflection (Renze and Guven 2024, Zhao et al. 2025), where the model revisits and revises earlier predictions, and curriculum reasoning (Maharana and Bansal 2022, Cornacchia and Mossel 2023), where it prioritizes simpler cases before tackling harder ones—both resembling strategies used by human experts. In summary, our key contributions include: (i) We introduce `CellPuzzles`, a novel benchmark that reformulates cell type annotation as a batch-level reasoning task, requiring the joint assignment of unique cell types to each cell based on both expression profiles and contextual metadata. (ii) We propose `Cell-o1`, a reasoning-enhanced LLM trained via SFT on distilled reasoning traces and RL with batch-level rewards, enabling interpretable annotation across cell batches. (iii) We conduct analyses of training dynamics, model behaviors, and error patterns, revealing emergent expert-like reasoning capabilities and robust performance on unseen biological conditions.

Related work

Traditional cell type annotation

Manual cell type annotation typically follows a multi-step process combining clustering, marker gene identification, and reference-based assignment. Cells are first grouped using unsupervised clustering algorithms such as hierarchical clustering, Louvain, or Leiden (Butler et al. 2018, Traag et al. 2019). Experts then inspect the resulting clusters, identify distinguishing marker genes, and assign cell type labels based on prior biological knowledge and curated reference databases (Aran et al. 2019, Stuart et al. 2019). The annotation process often involves iterative refinement, requiring the integration of various data sources and repeated evaluations. Although effective, expert-driven annotation is slow, labor-intensive, and requires substantial domain expertise. It also suffers from limited reproducibility and scalability when applied to large or multi-batch datasets (Abdelaal et al. 2019, Stuart et al. 2019). Moreover, its inability to systematically incorporate biological context further limits robustness and interpretability (Trapnell 2015).

Single-cell foundation models

Recent advancements in deep learning have led to the emergence of powerful single-cell foundation models designed to capture complex gene expression patterns through deep representation learning (Yang et al. 2022, Theodoris et al. 2023, Hao et al. 2024). Built upon transformer architectures, these models leverage extensive unsupervised pre-training on large-scale single-cell transcriptomic datasets to achieve enhanced representation quality and downstream task performance. These foundation models have demonstrated remarkable capabilities in transferring learned knowledge across different datasets and tasks, including cell type annotation (Yang et al. 2022, Theodoris et al. 2023), gene network inference (Osorio et al. 2020,

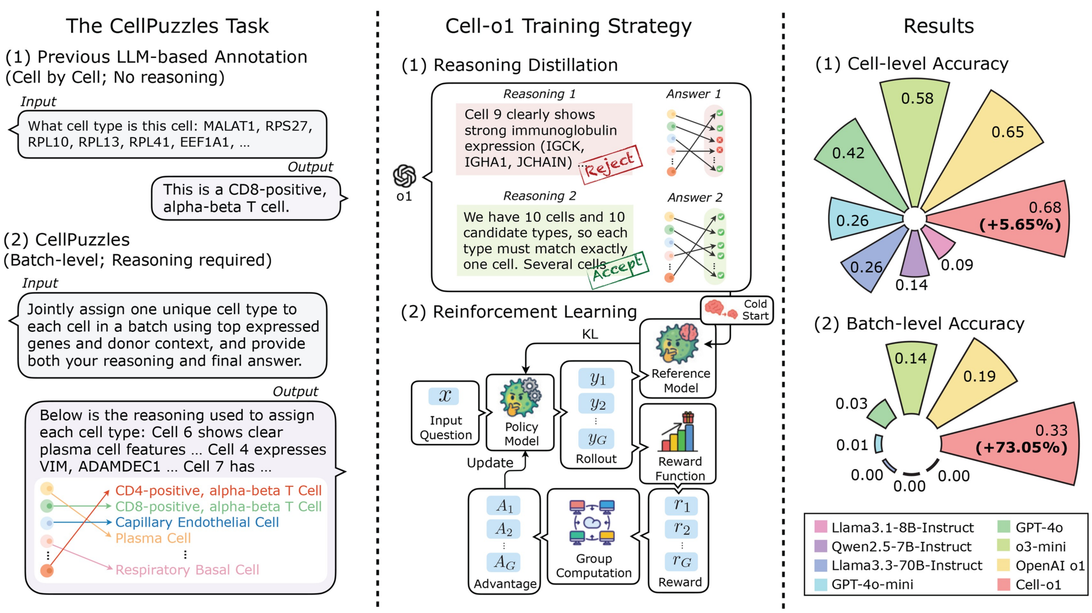

Figure 1 Overview of this work. `Cell-o1` achieves state-of-the-art on the `CellPuzzles` task.

Van de Sande et al. 2020), and perturbation prediction (Lotfollahi et al. 2019, Ji et al. 2021). Nevertheless, these models typically operate at the single-cell level, independently annotating each cell. Their performance in zero-shot settings remains limited, often necessitating fine-tuning with additional labeled data (Yang et al. 2022, Theodoris et al. 2023, Cui et al. 2024).

LLM-driven single-cell annotation

Recent research efforts have explored adapting single-cell gene expression data for LLMs, thereby leveraging their strong instruction-following, representation learning, and generalization capabilities (Achiam et al. 2023, Touvron et al. 2023). With their instruction-following and in-context learning capabilities, LLMs can quickly adapt to different single-cell tasks, an advantage that traditional single-cell analysis methods cannot achieve (Brown et al. 2020, Lou et al. 2024). To bridge the modality gap between scRNA-seq data and language models, several strategies have been proposed, including ranking genes by expression to form ordered lists (Hou and Ji 2024, Levine et al. 2024), embedding genes based on curated textual descriptions (Liu et al. 2023, Chen and Zou 2025), and directly encoding raw expression profiles using multimodal architectures (Schaefer et al. 2024, Zhao et al. 2024, Fang et al. 2025, Shi et al. 2025). These approaches enable LLMs to process biological information and perform tasks such as cell type annotation within a textual or multimodal framework. While promising, existing LLM-driven approaches still face challenges. Most approaches process cells individually, not fully exploiting batch-level context information. Moreover, they output annotation results without explicitly articulating the underlying reasoning process (Kojima et al. 2022, Wei et al. 2022), limiting interpretability and biological trustworthiness.

Methods

`CellPuzzles`: batch-level reasoning for cell type annotation

In real-world single-cell analysis, cell type annotation is rarely performed on individual cells in isolation. Instead, it typically occurs at the *batch level*, where groups of cells from the same donor or sample are jointly analysed. Experts commonly cluster cells based on gene expression profiles, identify representative marker genes for each cluster, and assign labels by integrating both expression patterns and contextual metadata such as tissue origin and disease status.

To emulate this expert-driven process, we introduce `CellPuzzles`, a novel benchmark shown in Fig. 2 that formulates cell type annotation as a batch-level reasoning task. Formally, each instance consists of a batch of N cells $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$, each sampled from a distinct cell type within the same donor and experimental batch. Each cell c_i serves as a proxy for a cluster centroid and, following (Levine et al. 2024), is represented by a ranked list of its top M expressed genes, $g_i = [g_{i1}, g_{i2}, \ldots, g_{iM}]$.

The entire batch is associated with a contextual description m, derived from donor-level metadata such as tissue type, disease status, sex, developmental stage, and other biologically relevant attributes. We also provide a candidate label set $\mathcal{Y} = \{y_1, y_2, \ldots, y_N\}$, which contains the ground-truth cell types for the N cells in the batch, randomly shuffled to remove positional biases. This setup mirrors the expert annotation scenario, where cell types are selected from a biologically plausible candidate pool rather than freely generated. The objective is to learn a mapping $f : \mathcal{C} \rightarrow \mathcal{Y}$ that assigns each cell to a unique label in $\mathcal{Y}$ by jointly considering gene expression profiles and contextual metadata, while generating interpretable reasoning traces to justify the label assignments.

As illustrated in Fig. 3, `CellPuzzles` is built from human single-cell datasets in the CELLxGENE portal (https://cellxgene.cziscience.com/) (Program et al. 2025) that span diverse tissues, disease conditions, and donor demographics. It comprises 8007 batches in total. Dataset construction details are in Supplementary Appendix *Dataset*.

`Cell-o1`: large reasoning model for cell batch annotation

Built upon `CellPuzzles`, we introduce `Cell-o1`, a large reasoning model built on the `Qwen2.5-7B-Instruct` backbone that emulates expert annotation strategies for batch-level

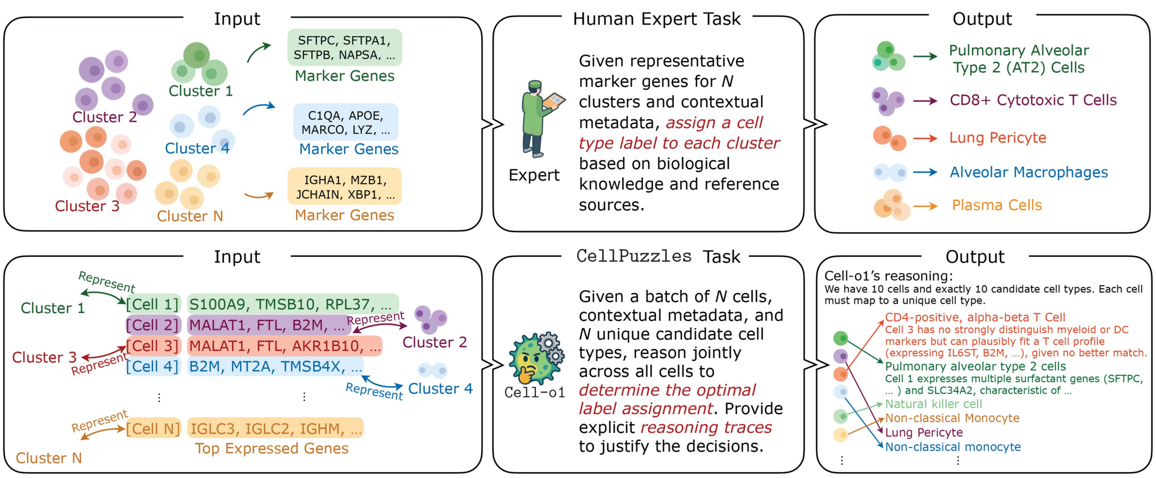

Figure 2 `CellPuzzles` formulates cell type annotation as a batch-level reasoning task that integrates gene expression and contextual metadata, inspired by how experts annotate cells in practice.

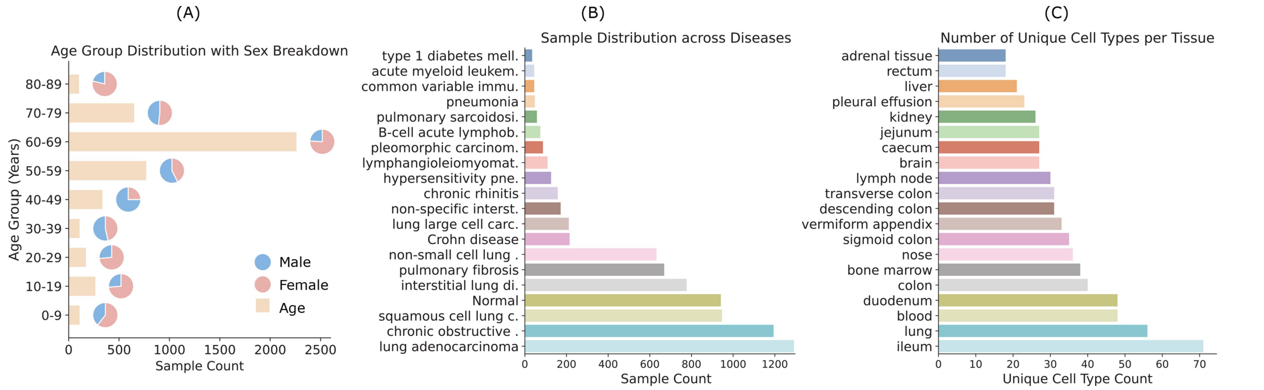

Figure 3 Data distribution of CellPuzzles. (A) Age group distribution with male-female breakdown. (B) Sample distribution across a wide range of disease conditions. (C) Number of unique cell types observed across different tissues. For clarity, only the top 20 entries are shown in (B) and (C).

single-cell analysis. It performs reasoning over top-expressed genes and shared biological context to assign unique cell type labels from a given candidate set.

Figure 1 illustrates our training strategy. We begin by introducing a unified prompt template to standardize model outputs and maintain consistent formatting across training stages. Next, we perform reasoning distillation via rejection sampling to construct a high-quality dataset of biological interpretations, followed by SFT for cold-start initialization. Finally, we apply RL with GRPO, guided by batch-level reward signals, to further refine the model.

Prompt template for structured reasoning

To ensure unified output formatting across training stages, we design a standardized prompt template that is used throughout distillation, SFT, and RL. As shown in Table 1, this template promotes global reasoning over the entire cell batch by instructing the model to integrate contextual information and jointly assign cell types to all cells from the candidate set.

Reasoning distillation and cold start

The batch-level reasoning task introduced in our benchmark poses significant challenges for model training. Unlike conventional classification tasks, the model must simultaneously analyse a group of cells, compare gene expression patterns, incorporate shared metadata, and generate a consistent label assignment. The high interdependence between inputs and outputs makes it difficult to produce valid and correct predictions, particularly at early training stages. We begin by performing reasoning distillation using OpenAI o1 (Jaech et al. 2024), a frontier LLM with strong multi-step reasoning capabilities, to construct a synthetic dataset of high-quality reasoning traces and predictions. This distilled dataset is then used for SFT, serving as a cold start initialization (Guo et al. 2025) for the subsequent RL.

Reasoning distillation via rejection sampling

We use OpenAI o1 (Jaech et al. 2024) to generate reasoning traces and predictions for 10,155 instances in CellPuzzles. Each input is paired with the standardized prompt shown in

Table 1 Prompt template guiding structured reasoning and prediction in Cell-o1.

You are an expert assistant specialized in cell type annotation. You will be given a batch of *N* cells from the same donor, where each cell represents a unique cell type. For each cell, the top-expressed genes are provided in descending order of expression. Using both the gene expression data and donor information, determine the correct cell type for each cell. You will also receive a list of *N* candidate cell types, and each candidate must be assigned to exactly one cell. Ensure that you consider all cells and candidate types together, rather than annotating each cell individually. Include your detailed reasoning within <think> and </think> tags, and provide your final answer within <answer> and </answer> tags. The final answer should be a single string listing the assigned cell types in order, separated by ' | '.

Table 1. For each instance, we generate 8 candidate responses. We then apply rejection sampling to filter out low-quality outputs. A response is accepted only if it (i) adheres to the expected format and (ii) yields a cell-type assignment that exactly matches the ground-truth labels. This process results in a distilled dataset containing 3912 accepted examples, corresponding to an acceptance rate of 38.52%.

The collected reasoning traces exhibit several desirable properties for guiding models toward expert annotation behavior and serve as a strong foundation for SFT: First, they reflect domain-specific knowledge by referencing known gene markers and biological associations. Second, they demonstrate global reasoning behavior, in which multiple cells and candidate labels are considered jointly to ensure consistency in label assignments. Third, the reasoning is structured and interpretable, disentangled from the final prediction and formatted in a standardized prompt-response style.

Supervised fine-tuning with distilled reasoning

We apply SFT on the distilled dataset to initialize the model before RL. While the final training objective involves optimizing reward signals, we find that starting from a purely pre-trained large language model fails to yield meaningful learning progress. Without prior exposure to the task-specific reasoning format and prediction structure, the model struggles to generate valid outputs. In practice, this leads to persistent formatting errors and incorrect label assignments during early-stage rollouts, resulting in zero or negative rewards and causing the policy to stagnate.

SFT serves as a cold start mechanism that mitigates this problem by providing the model with high-quality demonstrations of structured reasoning and correct predictions. By learning from the distilled dataset, the model acquires the basic capacity to follow instructions, adhere to the required response format, and reason across multiple entities within a batch. This improves response validity, reduces reward sparsity, and facilitates more stable and efficient policy optimization in the RL stage.

Reinforcement learning with GRPO

Group relative policy optimization

To reduce the training cost of RL, we adopt Group Relative Policy Optimization (GRPO) (Shao et al. 2024), which avoids training a separate critic by estimating advantages from a group of rollouts. Given a training input x, GRPO samples a group of G candidate responses $\{y_i\}_{i=1}^{G}$ from the old policy $\pi_{\theta_{\text{old}}}$. The updated policy π_θ is then optimized to maximize a clipped, advantage-weighted likelihood ratio objective, regularized by a KL divergence term that penalizes deviation from a reference policy $\pi_{\theta_{\text{ref}}}$. In this formulation, each candidate response contributes to the update based on its normalized advantage, and clipping is applied to ensure training stability. The KL penalty helps maintain proximity to a trusted reference policy. Further mathematical details of the GRPO objective are in Shao et al. (2024).

Reward modeling

To incentivize structured reasoning and ensure answer validity, we design a rule-based reward function that evaluates both the format and correctness of the model output. As instructed in Table 1, the model is required to generate responses in a strict format containing exactly one reasoning segment enclosed in <think>…</think> tags and one answer segment enclosed in <answer>…</answer> tags. Responses with incorrect tags, or extra texts, are considered invalid and receive a penalty.

For valid responses, we extract the predicted answer $\widehat{\mathbf{y}} = [\widehat{y}_1, \ldots, \widehat{y}_N]$ and compare it to the ground truth labels $\mathbf{y} = [y_1, \ldots, y_N]$. The batch-level reward is computed as:

$$R_{\text{batch}} = \begin{cases} \prod_{i=1}^{N} \mathbf{1}(\widehat{y}_i = y_i), & \text{if format is valid} \\ -1, & \text{if format is invalid} \end{cases} \quad (1)$$

Here, $\mathbf{1}(\cdot)$ is an indicator function: reward is 1 if all predictions are correct, else 0.

Experiments

Configuration

Dataset

We conduct experiments on the proposed `CellPuzzles`. Each instance comprises $8 \le N \le 15$ cells, each represented by its ranked top-expressed genes and a shared donor-level contextual description. For reasoning distillation, we use 10 155 CellPuzzles instances and retain 3912 accepted examples for supervised fine-tuning. We further sample 3000 additional instances for reinforcement learning, and use a held-out test set of 1095 batches for final evaluation. More details on the dataset processing can be found in Supplementary Appendix *Dataset*.

Evaluation metrics

We report results using the following metrics: (i) *Cell-level Accuracy:* The average proportion of correctly predicted labels per batch. (ii) *Batch-level Accuracy:* The proportion of batches where all predicted labels exactly match the ground truth. (iii) *Format Validity:* The proportion of outputs that follow the required response format. (iv) *Answer Uniqueness:* The average proportion of unique cell type predictions per batch.

Main results

Table 2 reports model performance on the `CellPuzzles` test set. We compare open-source LLMs [e.g. `Llama` (Touvron et al. 2023), `Qwen` (Bai et al. 2023)], closed-source LLMs with strong reasoning capabilities [e.g. `GPT-4o` (Achiam et al. 2023), OpenAI `o1`], and instruction-tuned baselines that directly predict answers from the input without producing reasoning traces. More details are in Appendix *Baselines*.

`Cell-o1` achieves the highest accuracy at both the cell and batch levels, outperforming all other models. Cell-level accuracy is consistently higher than batch-level accuracy across all models, which is expected given the stricter requirement of the latter. While it trails slightly behind OpenAI `o1` and `o3-mini` in format validity and answer uniqueness, `Cell-o1` still performs competitively on these dimensions, especially considering its smaller model size. Zero-shot models like `Llama3.1-8B-Instruct` often produce overly long outputs due to repeating cell labels or generating more predictions than required. Instruction-tuned variants without reasoning generate much shorter responses with better format adherence, but lack the ability to perform explicit, interpretable reasoning. Closed-source models such as `GPT-4o` and OpenAI `o1` show stronger instruction-following and reasoning consistency, producing well-structured outputs more reliably than most open-source models.

To further understand model behavior under varying conditions, we conduct a fine-grained analysis. Figure 4(A) shows that all models experience a decline in accuracy as the number of cells per batch increases, with `Cell-o1` consistently achieving higher scores across the full range. Figure 4(B) further shows that `Cell-o1` achieves the highest median and lowest variance in per-batch cell-level accuracy, indicating both strong and stable predictions across diverse contexts.

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Table 2 Evaluation of model performance on the `CellPuzzles` benchmark.

Model	Response length	Cell-level Acc	Batch-level Acc	Format	Uniqueness
Zero-shot performance					
`Llama3.1-8B-Instruct`	2264.9	0.0050	0.0000	0.0393	0.0190
`Qwen2.5-7B-Instruct`	879.8	0.1353	0.0000	0.9425	0.4528
`Llama3.3-70B-Instruct`	1006.8	0.2573	0.0064	0.9973	0.8321
`GPT-4o-mini`	779.9	0.2581	0.0027	0.9918	0.6565
`GPT-4o`	1087.0	0.4248	0.0283	0.9817	0.8179
`o3-mini`	1130.2	0.5804	0.1352	1.0000	1.0000
OpenAI o1	906.6	0.6479	0.1900	1.0000	0.9997
Instruction-tuning performance					
`Llama3.1-8B-Instruct`♠	71.9	0.6411	0.2969	0.9078	0.9071
`Qwen2.5-7B-Instruct`♠	70.9	0.6179	0.2475	0.9443	0.9404
Ours					
`Cell-o1`	1145.6	0.6849	0.3288	0.9826	0.9824

♠ indicates instruction-tuned baselines trained without any reasoning tasks, which directly predict answers from the input without generating reasoning traces. The best and second-best results are highlighted in red and blue, respectively.

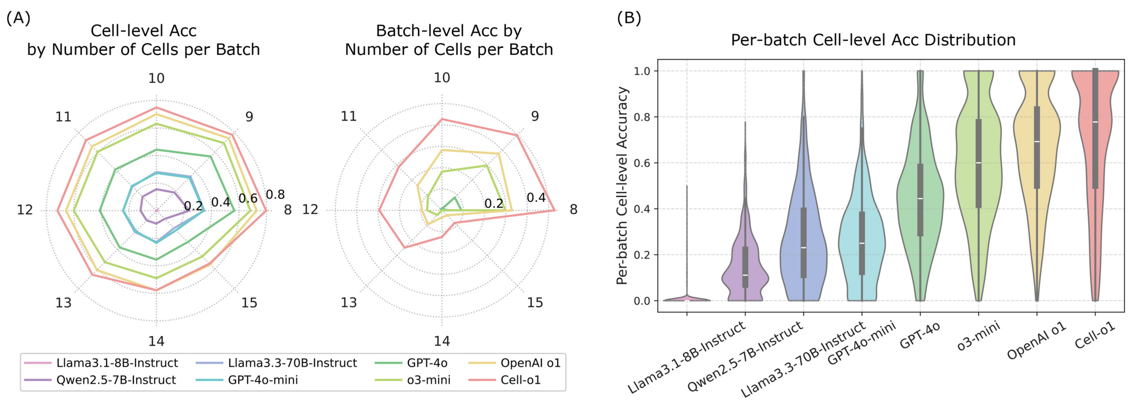

Figure 4 Fine-grained analysis of model performance. (A) Cell-level and batch-level accuracy across varying numbers of cells per batch. (B) Distribution of per-batch cell-level accuracy across models.

Generalization to unseen diseases

To evaluate the robustness and generalization ability of `Cell-o1`, we conduct evaluation on cell batches from disease types that were not seen during training. This setting simulates real-world scenarios where models are applied to new biological conditions without task-specific fine-tuning.

Specifically, we collect raw scRNA-seq data from four disease conditions—*Breast Cancer*, *Melanoma*, *Systemic Lupus Erythematosus (SLE)*, and *Colorectal Cancer*—and follow the `CellPuzzles` construction protocol to build corresponding test batches for zero-shot evaluation, resulting in a total of 539 annotated instances. As shown in Table 3, `Cell-o1` consistently outperforms all baselines across both cell- and batch-level accuracy. Closed-source models such as OpenAI `o1` exhibit strong generalization, but `Cell-o1` still leads by a significant margin despite its smaller model size. Compared to instruction-tuned models, which tend to make direct predictions without reasoning, `Cell-o1` generalizes more effectively to unseen disease conditions. This may be attributed to its ability to reason step-by-step during prediction, allowing it to adapt more flexibly to novel biological contexts.

Analysis and discussion

Training and inference dynamics under different configurations

To understand how different training strategies affect model performance and behavior, we compare four training configurations that differ along three dimensions: the presence of SFT initialization, the RL algorithm [GRPO vs. PPO (Proximal Policy Optimization) (Schulman et al. 2017)], and the design of the reward function.

As shown in Fig. 5, `Cell-o1` (A) with SFT initialization, GRPO optimization, and batch-level reward achieves stable training dynamics, controlled response length, and the best test performance. The response length fluctuates briefly at the beginning of training and then stabilizes, indicating that the model learns to balance output structure and reward optimization during training. Removing SFT initialization (B) leads to ineffective reward learning and a sharp increase in response length, indicating poor task alignment. Replacing GRPO with PPO (C) results in noisier reward trajectories and limited test performance gains, likely due to a mismatch between the reward objective and the

Table 3 Evaluation of model performance on unseen disease datasets.

Model	Breast cancer		Melanoma		SLE		Colorectal cancer		Average	
	Cell	Batch	Cell	Batch	Cell	Batch	Cell	Batch	Cell	Batch
Zero-shot performance										
Llama3.1-8B-Instruct	0.0075	0.0000	0.0179	0.0000	0.0035	0.0000	0.0147	0.0000	0.0109	0.0000
Qwen2.5-7B-Instruct	0.2382	0.0000	0.2679	0.0000	0.2400	0.0000	0.1996	0.0000	0.2364	0.0000
Llama3.3-70B-Instruct	0.4400	0.0000	0.5337	0.0159	0.3165	0.0100	0.4218	0.0000	0.4280	0.0065
GPT-4o-mini	0.4171	0.0000	0.5198	0.0000	0.4129	0.0000	0.4549	0.0000	0.4512	0.0000
GPT-4o	0.5847	0.0560	0.7619	0.1587	0.6576	0.1000	0.5715	0.0239	0.6439	0.0847
o3-mini	0.7806	0.2960	0.7341	0.1111	0.7706	0.3200	0.7208	0.1713	0.7515	0.2246
OpenAI o1	0.8136	0.3200	0.7738	0.1270	0.8200	0.3700	0.7845	0.3108	0.7980	0.2819
Instruction-tuning performance										
Llama3.1-8B-Instruct♠	0.7872	0.2880	0.7837	0.3333	0.7176	0.2500	0.7208	0.2590	0.7523	0.2826
Qwen2.5-7B-Instruct♠	0.7250	0.1600	0.7282	0.1746	0.5553	0.0300	0.5816	0.1036	0.6475	0.1170
Ours										
Cell-o1	0.8089	0.3360	0.8591	0.5397	0.8235	0.3800	0.7820	0.3028	0.8184	0.3896

♠ indicates instruction-tuned baselines trained without any reasoning tasks, which directly predict answers from the input without generating reasoning traces. The best and second-best results are highlighted in red and blue, respectively.

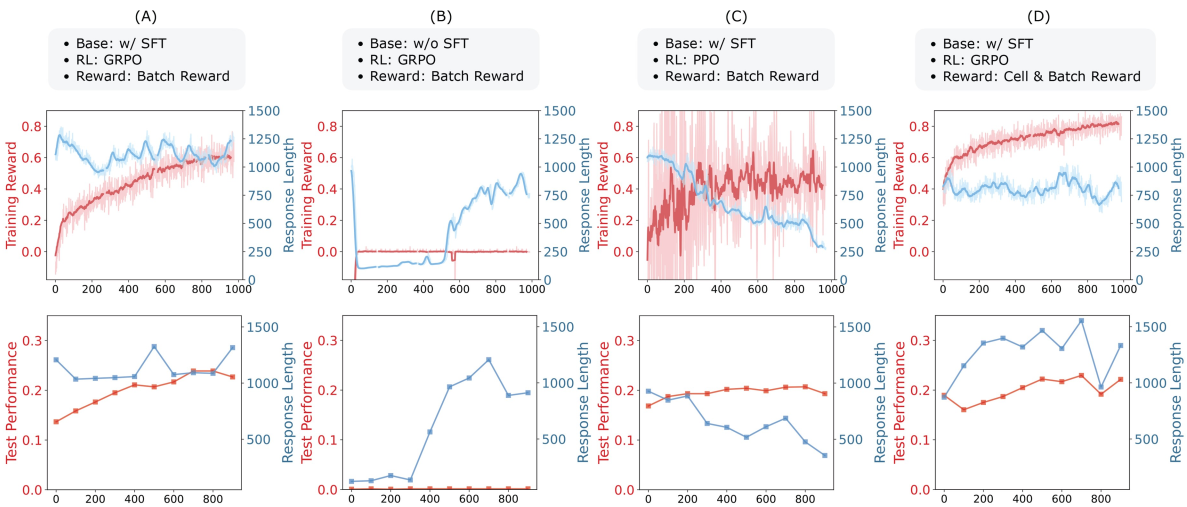

Figure 5 Comparing training and inference dynamics across different configurations.

final evaluation metric. The steadily declining response length suggests that PPO encourages shorter, less informative outputs to stabilize rewards, potentially at the expense of reasoning quality. Configuration (D) augments the reward with a cell-level accuracy component, using the average of cell and batch-level accuracy as the training signal. This leads to higher training reward scores, but slightly less stable test performance, which remains overall comparable to configuration (A).

Qualitative assessment of reasoning traces

To qualitatively assess the reasoning capabilities of our model, we conducted a human evaluation comparing its outputs against several baselines. We randomly sampled 100 test instances from CellPuzzles. For each instance, we constructed a step-by-step prediction diagram for each model, indicating the order in which cells were annotated (denoted by numbers), the predicted label, and whether each step was correct. If incorrect, experts further categorized the error into one of five predefined types (detailed in Supplementary Appendix *Experimental Setup Details*). Figure 6(A) shows one example.

To avoid bias, expert comparisons were conducted in a blind setting: for each instance, reasoning outputs from Cell-o1 and the baseline models were anonymized and presented in a randomized order. Experts then provided an overall judgment by comparing the reasoning quality across models, resulting in a win/tie/loss outcome for each pairwise comparison, as shown in Fig. 6(B).

During the human evaluation, we observed two interesting patterns exhibited by Cell-o1: self-reflection and curriculum reasoning. The former refers to the model's ability to revisit and revise its intermediate conclusions, while the latter reflects its tendency to tackle easier or more confident cases first before addressing more challenging ones. As shown in Fig. 6(C), Cell-o1 is the only model that consistently demonstrates both behaviors.

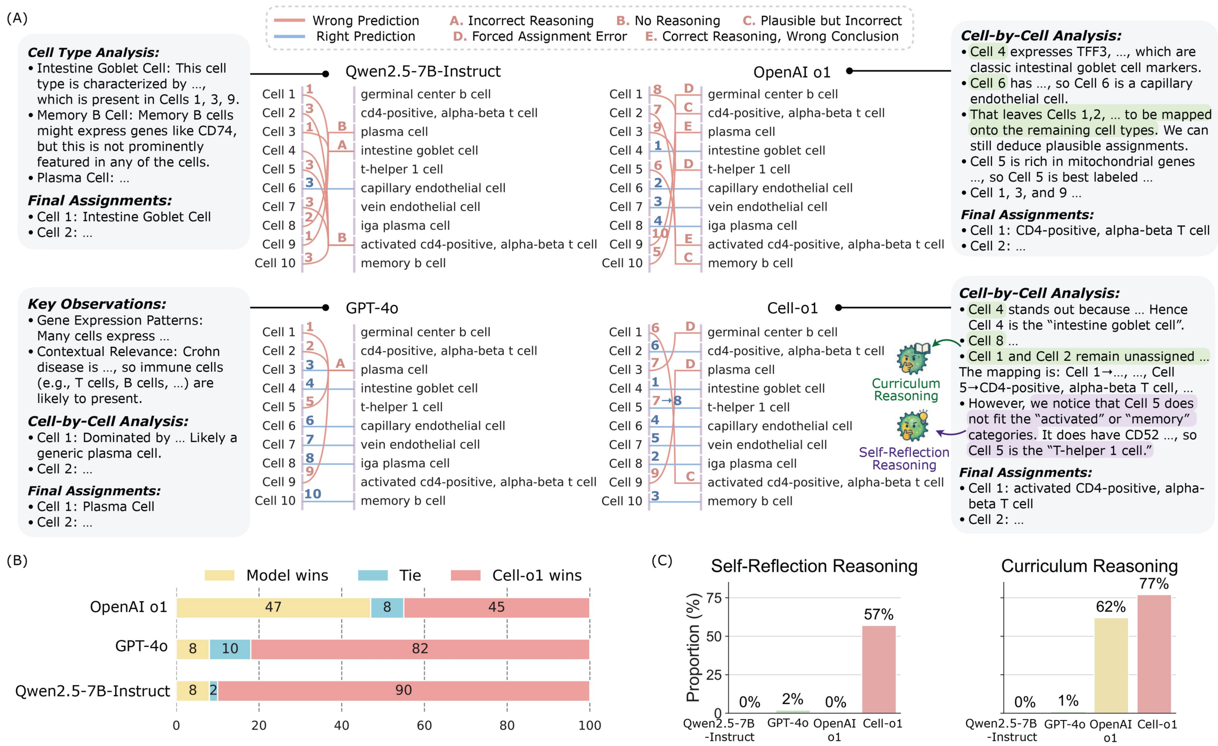

Figure 6 Analysis of model reasoning behavior. (A) Reasoning traces with steps numbered to reflect the model's reasoning order. Steps shown in blue represent correct predictions; those in red indicate incorrect predictions. (B) Expert evaluation of reasoning quality comparing `Cell-o1` with baseline models. (C) Proportion of traces exhibiting self-reflection and curriculum reasoning behaviors.

This highlights its ability to structure reasoning more deliberately—similar to how human experts approach complex annotation tasks.

Beyond the primary configurations analysed above, we compare cell-level vs. batch-level reasoning and open-ended vs. constrained QA settings in Appendix *Cell-level vs. Batch-level Reasoning* and *Open-ended QA vs. Constrained QA* to justify our final setup.

Conclusion

This work presents `CellPuzzles`, a structured reasoning benchmark for batch-level cell type annotation designed to emulate expert workflows. We further introduce `Cell-o1`, an LLM that demonstrates emergent expert-like reasoning behaviors, enabling accurate and interpretable predictions across diverse cellular contexts. Looking ahead, future directions include incorporating retrieval-augmented methods (Chen et al. 2025, Jiang et al. 2025, Jin et al. 2025, Xiong et al. 2025) to integrate external knowledge, leveraging domain ontologies (Ashburner et al. 2000) for more consistent reasoning, and extending structured reasoning to broader biomedical applications (Tian et al. 2023).

Limitations and broader impacts

While `Cell-o1` demonstrates strong batch-level reasoning for scRNA-seq cell type annotation, its evaluation is conducted under a controlled benchmark setting. Specifically, `CellPuzzles` constructs each instance using representative cells from distinct known cell types together with a matched candidate set. In real-world annotation settings, however, researchers do not know the candidate pool a priori, common cell types may appear multiple times, rare or novel cell types may be absent from the candidate set, and performance may depend on upstream clustering or representative-cell selection. Therefore, the current benchmark should be interpreted as evaluating batch-level reasoning under a simplified but informative setting rather than a complete end-to-end annotation workflow. In addition, `Cell-o1` relies on predefined candidate label sets, which limits its applicability to novel cell types. We also represent each cell using its top-*M* expressed genes rather than curated marker genes or DEGs. While this provides a simple and uniform input format across datasets, marker-based representations may better align with biological annotation practice and could affect model performance. `Cell-o1` was trained solely on Cell × Gene data, so its generalizability to other modalities or under-represented tissues remains untested. Additionally, `Cell-o1` depends on sparse and high-variance RL rewards that require careful hyperparameter tuning, may produce plausible but incorrect reasoning, and incur non-trivial computational overhead.

Nevertheless, by providing interpretable, group-aware predictions, `Cell-o1` can accelerate annotation pipelines and democratize access to high-quality labels, which potentially fuels discoveries in immunology, oncology, and developmental biology. On the other hand, it also poses risks that misapplication could propagate errors into downstream analyses, and large-scale pre-training and fine-tuning carry environmental and economic costs, underscoring the need for responsible, human-in-the-loop deployment and careful consideration of data privacy and dual-use concerns.

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Author contributions

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Supplementary material

Supplementary material is available at *Bioinformatics* online.

Conflict of interests

None declared.

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Data availability statement

Code and data are available at https://github.com/ncbi-nlp/cell-o1.

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